
5 Research Questions

New Description

Outcome	Weight
Explore of IT Areas	◆◆◆◆

Outcome	Weight
Context of Research Project	◆◆◆◆◇

Outcome	Weight
Communication of Research Outcomes	◆◆◆◆

Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆

Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆

Date	Author	Comment
2025/08/16 08:26	N/a Nikhil	Ready to Mark
2025/08/19 11:43	N/a Nikhil	Not Started
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DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Research Questions

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Outcome	Weight
Explore of IT Areas	◆◆◆◆◆
Context of Research Project	◆◆◆◆◇
Communication of Research Outcomes	◆◆◆◆◆
Validation of Objectives and Methodology	◆◆◆◆◆
Application of Skills and Knowledge	◆◆◆◆◆

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August 19, 2025



Research Questions Submission

Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation

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August 19, 2025

Research Questions

Based on the comprehensive literature review conducted, three critical research questions have been formulated to address identified gaps in GPS spoofing detection for autonomous mobile robots.

Research Question 1

What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?

Motivation: Current literature identifies fundamental weaknesses in civilian GPS signals but lacks comprehensive analysis specific to mobile robotic platforms [1,2]. Understanding these vulnerabilities is essential for developing targeted detection mechanisms.

Research Gap: Limited characterization of platform-specific vulnerabilities in resource-constrained mobile robots compared to aviation systems.

Research Question 2

How can sensor fusion techniques effectively detect GPS spoofing in real-time?

Motivation: Existing sensor fusion approaches show promise but lack implementation specifics for real-time operation on embedded platforms [3,4]. Cross-validation between GPS, IMU, and odometry requires optimization for mobile robots.

Research Gap: Insufficient real-time validation of sensor fusion methods in practical robotic navigation scenarios.

Research Question 3

How can a dynamic GPS Trust Index be designed to quantify the reliability of GPS data?

Motivation: Current detection methods provide binary outputs (spoofed/authentic) but lack quantitative reliability assessment [5]. A dynamic trust index enables gradual response to varying threat levels.

Research Gap: Absence of standardized trust quantification methods for continuous GPS reliability assessment in autonomous systems.

Conceptual Framework

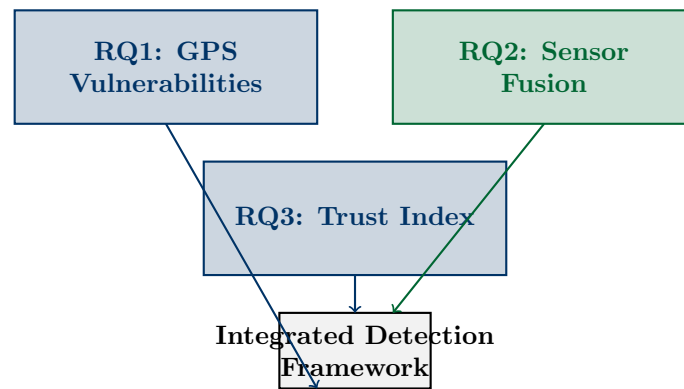


Figure 1: Research questions integration framework for GPS spoofing detection

Literature Theme Associations

- **RQ1** aligns with *Theme 1: GPS Vulnerabilities and Attack Taxonomies*
- **RQ2** corresponds to *Theme 2: Detection Techniques* and *Theme 3: Simulation Frameworks*
- **RQ3** relates to *Theme 4: Mitigation and Recovery Strategies*

Expected Contributions

This research will contribute to the field by:

1. Providing comprehensive vulnerability analysis specific to mobile robotic platforms
2. Developing practical sensor cross-validation algorithms for real-time implementation
3. Introducing a novel GPS Trust Index for dynamic reliability assessment
4. Establishing a reproducible simulation framework using ROS-Gazebo-PX4

References

- [1] T. E. Humphreys et al., "Assessing the spoofing threat: Development of a portable GPS civilian spoofer," *Proc. ION GNSS*, 2008.
- [2] A. J. Kerns et al., "Unmanned aircraft capture and control via GPS spoofing," *J. Field Robot.*, vol. 31, no. 4, pp. 617-636, 2014.
- [3] A. Cavaleri et al., "Detection of spoofed GPS signals at code and carrier tracking level," *Proc. IEEE/ION PLANS*, 2010.
- [4] J. Magiera and R. Katulski, "Detection and mitigation of GPS spoofing based on antenna array processing," *J. Appl. Res. Technol.*, vol. 13, no. 1, pp. 45-57, 2015.
- [5] M. Nayfeh, "Artificial intelligence-based GPS spoofing detection and implementation with applications to unmanned aerial vehicles," Ph.D. dissertation, Purdue Univ., 2023.